TT-L-57C
October 10, 1972
SUPERSEDING
Fed. Spec. TT-L-57B
May 12, 1961

FEDERAL SPECIFICATION

LACQUER, RUBBING, CLEAR, (FOR WOOD FURNITURE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of cellulose nitrate rubbing lacquer for wood furniture use. The material is available in gloss and flat finish.

1.2 Classification.

1.2.1 Types and classes. The lacquer shall be of the following types and classes.

Type I - Ready to spray (cold application).
Class 1 - Gloss
2 - Flat

Type II - For hot application.
Class 1 - Gloss
2 - Flat

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-F-336 - Filler, Wood, Paste.
TT-N-350 - Nitrocellulose, Technical (For Use in Organic Coatings).
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-S-190 - Sealer, Sanding, Lacquer Type (For Wood Furniture).
TT-S-300 - Shellac, Cut.
TT-S-720 - Stain, Wood, Nongrain Raising, Solvent-Dye Type.
TT-T-266 - Thinner; Dope and Lacquer (Cellulose-Nitrate).
TT-T-291 - Thinner, Paint, Volatile Spirits, Petroleum Spirits.
CCC-C-419 - Cloth, Cotton Duck, Unbleached, Plied Yarns (Army and Numbered).

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a aprt of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Forest Products Laboratory Wood-Handbook

United States Department of Commerce Publication:
Product Standard PS-51-71; Hard and Decorative Plywood

(Application for copies should be addressed to the Superintendent of Docuemtns, Government Printing Office, Washington, DC 20402. Price of handbook, 75 cents; price of product standard; 10 cents.)

American Society for Testing and Materials (ASTM) Standards:

D 1211 - Method of Test for Temperature Change Resistance of Clear Nitrocellulose Lacquer Films Apply to Wood.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The lacquer covered by this specification shall contain nitrocellulose conforming to TT-N-350, type II processed to conform to all the requirements specified herein. The thinner used in the process shall comply to "Rule 66"1/. A certified statement to this effect is necessary.

3.2 Qualitative requirements.

3.2.1 Condition in container. The freshly opened container shall show no deterioration of the can and when tested as in 4.3.1 the lacquer shall be free of foreign matter such as skin, dirt, or undissolved ingredients.

3.2.2 Storage stability.

3.2.2.1 Partially full container. When tested as in 4.3.2.1 the lacquer shall show no skinning. After storage as specified in 4.3.2.1 it shall show no curdling, skinning, and gummy sediment. It shall remain in a homogeneous state.

3.2.2.2 Full container. The lacquer when stored for 12 months as tested in 4.3.2.2 shall show no skinning, caking, curdling and gummy sediment. The lacquer shall be useable with no appreciable change in viscosity and drying time.

3.2.3 Color. The color of types I and II lacquer, when reduced as specified in 4.3.3, shall be no darker than a solution of 0.384 gram of reagent-grade potassium dichromate in 100 milliliters of concentrated sulfuric acid (Sp. gr. 1.84) or the No. 7 tube in accordance with method 4248 of Fed. Test Method Std. No. 141.

3.2.4 Working properties. The lacquer shall spray satisfactorily in all respect and shall show no running, or sagging when tested as in 4.3.4. The appearance of the dried film shall show no dusting, mottling, seeding, wrinkling, pitting, or blistering, open pores, pinholes or film fractures and no more than a trace of orange-peeling.

3.2.5 Rubbing properties. When tested as in 4.3.5 the film (after rubbing) shall have a smooth finish that show no sweatback.

3.2.6 Self-lifting. The lacquer shall show no tendency of self-lifting when tested as in 4.3.6.

1/ Information on Rule 66 may be obtained from The Los Angeles Air Pollution Control District Los Angeles, CA 90013.

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3.2.7 Resistance tests:

3.2.7.1 Alcohol. When tested as in 4.3.7.1 any whitening or spotting that develops must be easily removed with light polishing.

3.2.7.2 Naphtha. There shall be no discoloration nor marring that cannot be easily removed with light polishing when tested as in 4.3.7.2.

3.2.7.3 Boiling water. The lacquer film when tested as in 4.3.7.3 shall show no graying or spotting.

3.2.7.4 Cold check test. When tested as in 4.3.7.4 there shall show no checking or cracking at the end of an exposure of ten cycles.

3.2.7.5 Print resistance. Any imprint that occurs in either cold or hot tests shall readily be removed with liquid polish when tested as in 4.3.8.

3.2.7.6 Ultra-violet light resistance. There shall be no more than a slight darkening of the lacquer film at the end of 24 hours exposure when tested as in 4.3.9.

3.2.8 Adhesion and toughness. The lacquer film when tested as in 4.3.10 shall not crack and flake. The film shall adhere to the substrate.

3.3 Quantitative requirements.

3.3.1 The lacquer shall meet the quantitative requirements specified in table I.

TABLE I. Quantitative requirements

Characteristics	Requirements Minimum	Maximum
Nonvolatile matter, percent by weight:		
Type I	21	---
Type II	30	---
Viscosity, stokes		
Type I	0.5	1.0
Type II	0.5	1.65
Dry to rub:		
Type I (hour)	---	18
Type II (hour)	---	18
Benzol:		
Type I	Negative	---
Type II	Negative	---

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be performed in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Test procedures. The tests shall be conducted in accordance with test methods of Fed. Test Method Std. No. 141, except as otherwise specified in table II.

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TABLE II. Index

Characteristics	Requirements Reference	Applicable Test	
		Fed. Test Method Std. No. 141	Paragraph Reference
Nonvolatile matter	Table I	4044	-------
Viscosity	Table I	4271	-------
Dry to rub	Table I	----	4.3.5
Benzol	Table I	5091	-------
Condition in container	3.2.1	----	4.3.1
Storage stability	3.2.2	----	4.3.2
Color	3.2.3	----	4.3.3
Working properties	3.2.4	----	4.3.4
Rubbing properties	3.2.5	----	4.3.5
Self-lifting	3.2.6	----	4.3.6
Resistance to alcohol	3.2.7.1	----	4.3.7.1
Resistance to naphtha	3.2.7.2	----	4.3.7.2
Resistance to boiling water	3.2.7.3	----	4.3.7.3
Cold check	3.2.7.4	----	4.3.7.4
Print resistance	3.2.7.5	----	4.3.8
Ultra-violet light	3.2.7.6	----	4.3.9
Adhesion and toughness	3.2.8	----	4.3.10

4.3.1 Condition in container. Determine package condition of the lacquer in accordance with method 3011.1 of Fed. Test Method Std. No. 141 for compliance with 3.2.1.

4.3.2 Storage stability.

4.3.2.1 Partially full container. Determine skinning after 48 hours in accordance with method 4141 of Fed. Test Method Std. No. 141 for compliance with 3.2.2.1.

4.3.2.2 Full container. In accordance with method 4142 of Fed. Test Method Std. No. 141, allow a full standard lined quart can of the lacquer to stand undisturbed for 12 months then evaluate the content for compliance with 3.2.2.2.

4.3.3 Color (alternate method). Using method 4241 of Fed. Test Method Std. No. 141 (except ruduce the solids content for type II lacquer to 20 to 21 percent by weight with thinner conforming to TT-T-266). Method 4242 or 4248 may be used as alternate methods. Observe for compliance with 3.2.3.

4.3.4 Working properties.

4.3.4.1 Preparation of panels (five panels). The panels shall be cut from synthetic resin-bonded straight grain, 5-ply walnut veneer (faces and backs) having no sizing or other surface pretreatment conforming to applicable requirements of Product Standard PS-51-71; Hard and Decorative Plywood. They shall be approximately 5/16 inch thick; 4-inches long and having a minimum area of 32 inch square. The panel shall have a moisture content of not less than 6.0 percent, and not more than 8 percent at the time of testing. Panels can be conditioned to a moisture content of 7 to 8 percent, under any of the following conditions (taken from the Forest Products Laboratory Wood Handbook):

65° to 75°F., and 35 to 40 percent relative humidity.
or
90° to 100°F., and 40 to 45 percent relative humidity.
or
120° to 125°F., and 45 to 50 percent relative humidity.

Prior to finishing, the panel shall be thoroughly sanded smooth with 5/0 or finer sandpaper. Glue-size may be used before this sanding to stiffen the wood fibers. Surfaces shall be thoroughly cleaned before staining.

4.3.4.2 Application of the system.

4.3.4.2.1 Staining. Apply a non-grain-raising "fast to light" solvent type stain conforming to TT-S-720 and allow to dry thoroughly for not less than 30 minutes.

4.3.4.2.2 Washcoat application. A washcoat consisting of a blend of seven parts denatured alcohol and one part shellac conforming to TT-S-300, type I, grade B, body 1, shall be applied. After the washcoat has thoroughly dried for not less than 30 minutes, the surface shall be scuff sanded with 5/0 or finer sandpaper, and all loose particles and dust removed.

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4.3.4.2.3 Filler. A paste wood filler conforming to TT-F-336 shall be applied. The filler shall be thinned with mineral spirits conforming to TT-T-291, type II, grade A, to increase its workability. When the filler has dried to a proper wiping consistency, it shall be worked into the pores by any means in such a way that all pores are completely filled. Excess filler shall be thoroughly wiped off the surface then allowed to air at room temperature for at least 12 hours.

4.3.4.2.4 Sealer. Apply a coat of sealer conforming to TT-S-190 and allow to dry for one hour then sand smooth with 5/0 or finer sandpaper.

4.3.4.2.5 Lacquer application.

4.3.4.2.5.1 Type I (cold lacquer). Apply two coats of lacquer on one of the panels prepared in 4.3.4.1 allowing one hour air-drying between coats. Allow 16 hours drying of the last coat before rubbing. Observe for compliance with 3.2.4.

4.3.4.2.5.2 Type II (hot lacquer). Using Underwriters Laboratory's hot spray method apply one cross coat of hot lacquer (see table III) on another panel prepared in 4.3.4.1. Allow to dry for 16 hours. Observe for compliance with 3.2.4.

TABLE III. Requirements for hot spray application

Atomization air temperature	Room temperature 70° to 90°F.
Atomization air pressure	60 - 65 pounds
Cold lacquer feed tank pressure	10 - 15 pounds
Fluid and air adjustment valves on gun	Wide open (DeVilbiss or equal type E tip & No. 765 air cap)
Air cap and fluid tip lacquer temperature at spray gun	140° - 150°F.
Distance of spray gun to work	7 - 8 inches
Lacquer	15 - 20 fluid oz. per minute
Thermostatic setting on lacquer heating unit	170°F. maximum

4.3.5 Rubbing properties. After final coat on both panels has dried 16 hours the lacquer film shall be sanded with No. 320 "wet or dry" sandpaper wetted and lubricated with mineral spirits conforming to TT-T-291, type II, grade A. The panels shall then be rubbed with pumice and oil or any suitable rubbing compound to a smooth finish. Clean the surface. Care shall be exercised not to cut through the finish on high spots. Examine the film immediately after rubbing and cleaning and again after 24 hours for conformance with 3.2.5.

4.3.6 Self-lifting properties. After the first coat of the lacquer has been applied as described in 4.3.4.2.5, apply a second, third, and fourth coats as follows after one hour, six hours and 24 hours respectively. Examine the film for compliance with 3.2.6.

4.3.7 Resistance tests (spot test). Allow the lacquer film as prepared in 4.3.4.2.5 to age for one week, then run the following tests.

4.3.7.1 Alcohol. Place 0.5 ml. of 100 proof (50 percent) grain neutral spirits on the aged panel and trap for 2 hours with a watch glass. Remove glass and allow to evaporate, lightly polish the exposed area using a soft cloth then examine for compliance with 3.2.7.1.

4.3.7.2 Naphtha. Trap 1 ml of naphtha (VM&P) on the lacquer film on the same panel as in 4.3.7.1 (another spot) for 24 hours, remove and wipe clean. Allow to dry for 2 hours, polish the exposed area with a soft cloth and examine for compliance with 3.2.7.2

4.3.7.3 Boiling water. Prepare a finished panel, as specified in 4.3.4.2.5. Pour 25 ml of boiling water on the leveled panel and allow to cool at room temperature. Dry off the surface of the lacquer coating and examine for compliance with 3.2.7.3.

4.3.7.4 Cold check resistance. This test procedure is intended to be the same as ASTM D 1211.

4.3.7.4.1 Preparation of panels. Three panels, prepared and finished as in 4.3.4.2.5, shall be aged for 10 days at not less than 70°F , and not more than 100°F.

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4.3.7.4.3 Evaluation. Veneer checks, or checks running parallel to the grain shall not be considered as failure, but the affected panel(s) shall be discarded and another panel shall be prepared and tested in lieu thereof. Grain cracking is to be regarded as poor panel preparation or glue line failure and tests repeated. True lacquer checking ordinarily shows up as one or more wavy or straight lines which cross the grain and do not seem to be related to grain structure. Any well defined lacquer checking answering this description, regardless of size, shall constitute a lacquer failure.

4.3.8 Print resistance tests. The panels finished as specified in 4.3.4.2.5 shall be allowed to air dry 48 hours after rubbing as in 4.3.4.2.5.2 then run the following tests:

4.3.8.1 Cold print. A piece (approximately 2 by 2 inches) of shelter tent Army duck conforming to CCC-C-419, type III, 8.25 ounces per square yard minimum count 60 x 58 yards per inch, shall be placed on the panel. A metal weight weighing one pound is place on the piece of duck. The contact surfaces of this weight shall be a smooth plane surface, circular in form, and one square inch in area. The weight shall remain in position for a period of 24 hours at 75°F. Immediately after removal of the weight, rub lightly the exposed areas with liquid polish on a pad of soft cloth then examine for compliance with 3.2.7.5.

4.3.8.2 Hot print. The same procedure shall be used for the hot print as is used for cold print except for the following: (a) The weight shall be 1/2 pound instead of 1 pound.
(b) The temperature during the pressure period shall be 110°F. instead of 75°F.

4.3.9 Ultra-violet light resistance. Spray 1/2 of a 6 x 9 inch white porcelain panel with the lacquer under test. For type I lacquer spray two coats with one hour drying time between coats. For type II lacquer shall be applied as one cross coat sprayed hot. The lacquer is allowed to air-dry a minimum of 18 hours. Expose the panel to a carbon arc, place 12 inches from the panel and enclosed by a Corex D glass globe, for a period of 24 hours. After exposure spray the other half of the panel in the same manner. Allow lacquer to air-dry a minimum of 4 hours. Compare the exposed lacquer with the unexposed one to determine compliance with 3.2.7.6.

4.3.10 Toughness and adhesion. A panel prepared as in 4.3.4.1 and rubbed as in 4.3.4.2.5.2 shall be forced dried in an oven at 125° ± 4°F. for 24 hours before testing. A steel rod, 5 inches long and 5/16 inches in diameter, with one end rounded and polished to a 5/32 inch spherical radius, shall be used. A mar is made on the panel by drawing the rounded end of the rod (hold perpendicular relative to the panel) over the finished surface. Enough pressure shall be exerted on the steel rod to produce an indentation at least 3/32 inch wide. The panel shall be marred both parallel and perpendicular to the wood grain. The indentation shall show no whitening which is evident of film separation. The film shall conform to the contour of the indentation and shall not crack in a brittle manner in or along this indentation. Determine for compliance with 3.2.8.

4.3.11 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined and tested to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The lacquer shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging and the level of packing shall be A, B, or C, as specified (see 6.2). The lacquer shall be furnished in 1-pint can, 1-quart can, 1-gallon can, and 5-gallon steel pails (see 6.2).

5.1.1 Precautionary markings. In addition to the markings required by TT-P-143, all individual containers should have the following marking:

CAUTION: Adequate precautions should be taken when spraying.

5.2 Special marking.

5.2.1 Each container shall contain manufacturer's instructions:

(a) Preparation of surface.
(b) Instruction for application.

6. NOTES

6.1 Intended use. The lacquer covered by this specification is intended for use on wood furniture.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Color required (see 3.2).
(d) Size of container required (see 5.1).
(e) Levels of packaging and packing required (see 5.1).

6.3 Basis of purchase. The lacquer covered by this specification should be purchase by volume, 231 cubic inches at 20°C (68°F.), equals one gallon.

6.4 Supersession data. Type I of this specification supersedes Military Specification MIL-L-1137c(QMC), Lacquer, Rubbing Clear (For Wood Furniture).

MILITARY CUSTODIANS:

Navy - YD
Air Force - 85

Review Activity:

Air Force - 85
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Military coordinating activities:

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